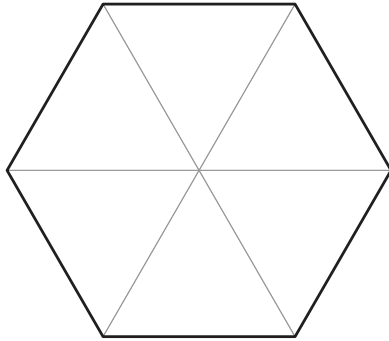
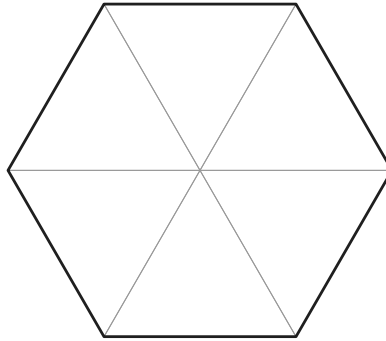


Problem 1: Cover hexagon A with green triangles only. Count the triangles. Write the number in the blank. Cover hexagon B with red trapezoids only and count them. Cover hexagon C with blue rhombi only and count them.



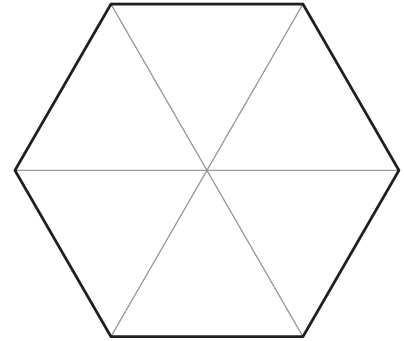
A

_____ green triangles



B

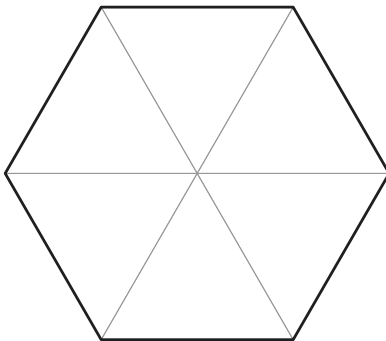
_____ red trapezoids



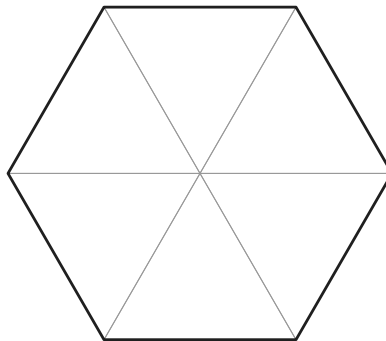
C

_____ blue rhombi

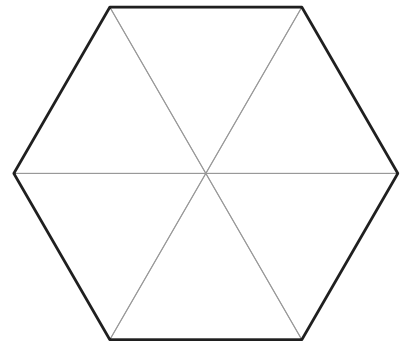
Problem 2: Cover the hexagon with blue rhombi. Color the first picture to show your way: color the two triangles under each rhombus with the same crayon, and use a different crayon for each rhombus. Find a different way and color it in the second picture. Look for a third way that is not the same as the first two. If you find one, color it in the third picture.



first way

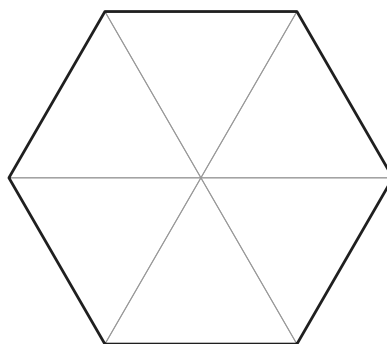
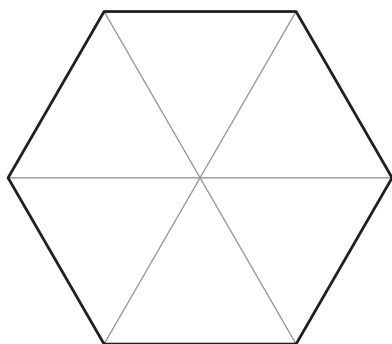
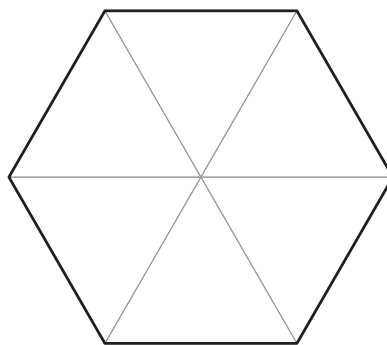
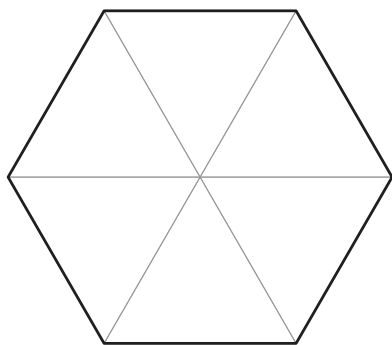


second way



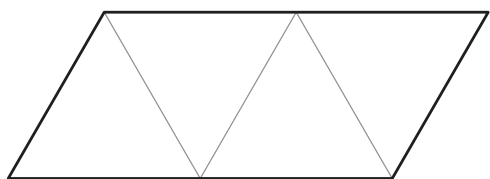
third way

Problem 3: Cover the hexagon with red trapezoids only. Color each trapezoid with its own crayon. Find every different way, and color each way in a new picture. You may not need all four pictures. Then cover the hexagon with green triangles only and find every different way. Write how many ways you found.



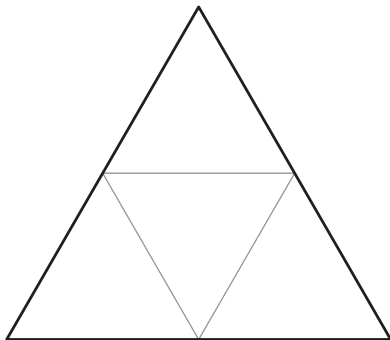
Ways with red trapezoids: _____ Ways with green triangles: _____

Problem 4: Try to cover each shape with blue rhombi only. A rhombus must lie exactly on two triangles of the picture, and rhombi may not overlap. If you cover the whole shape, draw a check in the box. If you cannot after three tries, draw an X in the box.



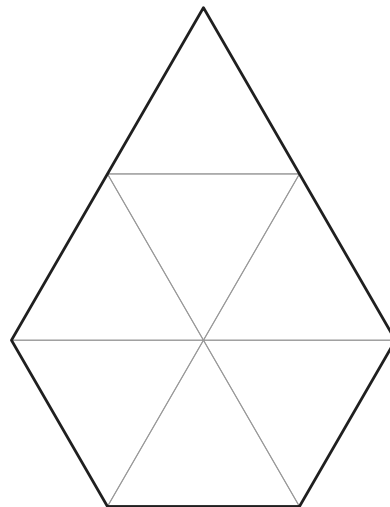
A ☐

up _____ down _____



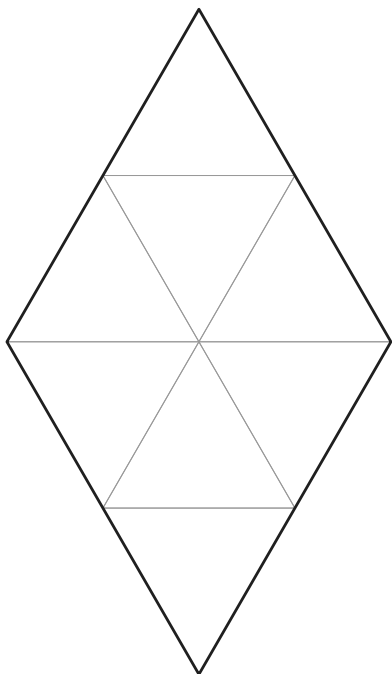
B ☐

up _____ down _____



C ☐

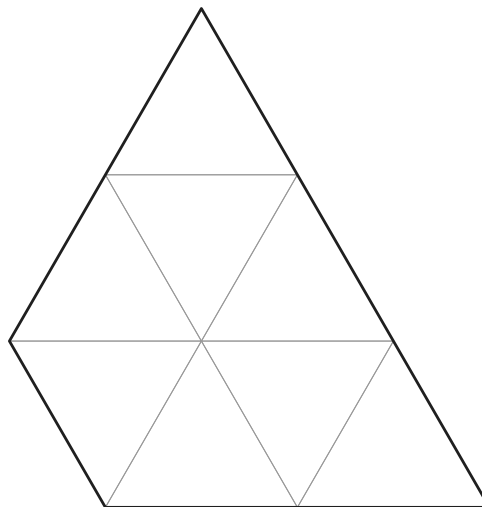
up _____ down _____



D



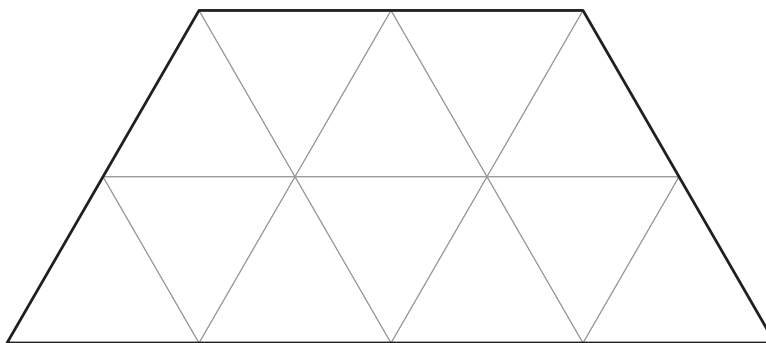
up _____ down _____



E



up _____ down _____



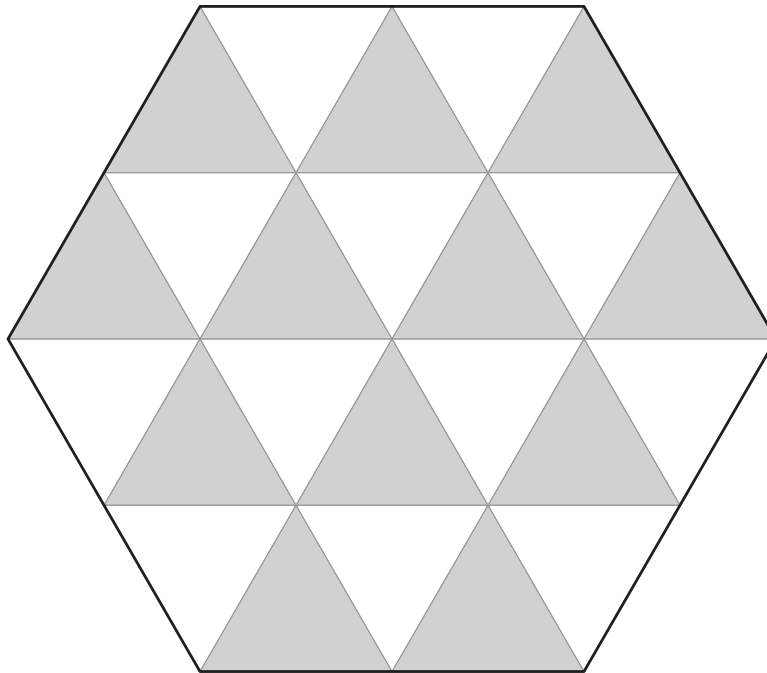
F



up _____ down _____

Problem 5: Cover each shape A to F with green triangles only. Count the triangles that point up and the triangles that point down. Write the two numbers under the shape.

Problem 6: In this picture the triangles that point up are gray and the triangles that point down are white. Put one blue rhombus anywhere on the picture so that it lies exactly on two triangles. Count the gray triangles under it and the white triangles under it. Write the two numbers in the first row. Move the rhombus to a new place, turned any way you like, and count again. Do this five times.



1. gray _____ white _____

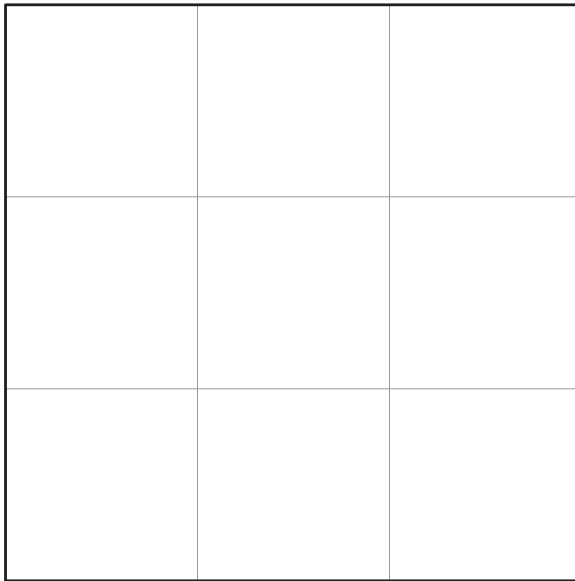
2. gray _____ white _____

3. gray _____ white _____

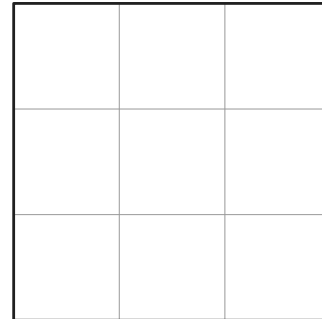
4. gray _____ white _____

5. gray _____ white _____

Problem 7: A domino covers two squares that share a side. Put a black square on one square of the big board. Cover all the other squares with dominoes. If you cover them all, draw a check on that square of the small map. If you cannot, draw an X there. Do this for every one of the nine squares.

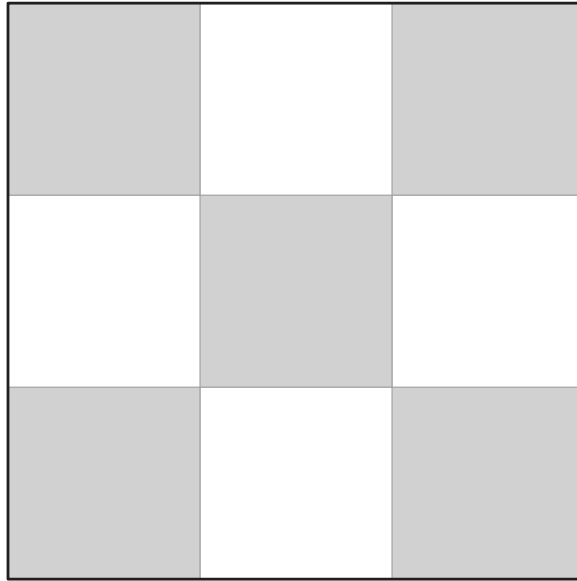


big board



small map

Problem 8: On this board the dark squares and the light squares make a checkerboard. Count the dark squares. Count the light squares. Write the two numbers. Then put one domino anywhere on the board. Count the dark squares under it and the light squares under it. Write the numbers. Move the domino and count again. Do this three times. Then look at your map from Problem 7: point to the squares with a check and say what color they have on this board.



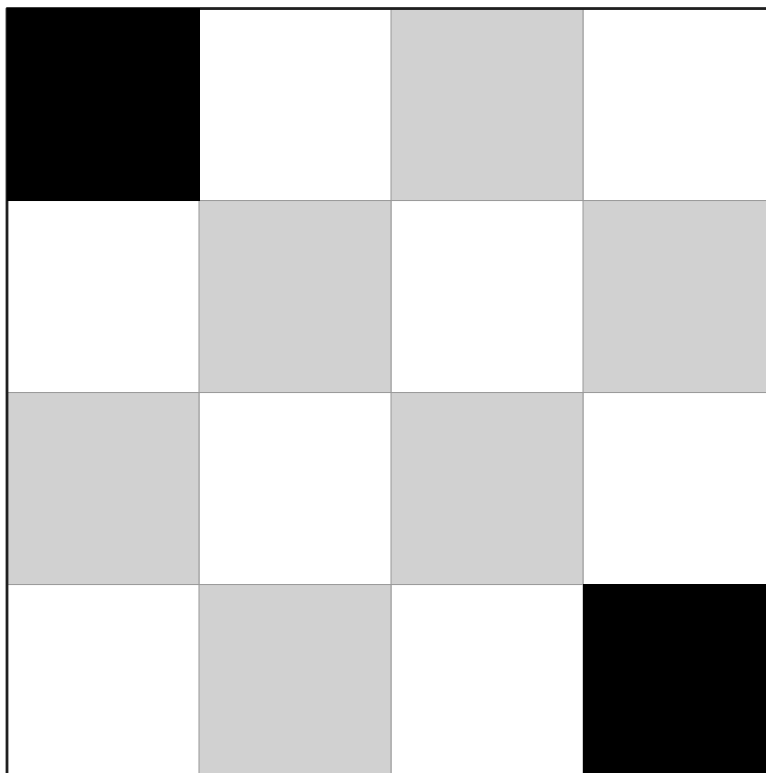
Dark squares on the board: _____ Light squares on the board: _____

Domino 1 covers dark _____ and light _____

Domino 2 covers dark _____ and light _____

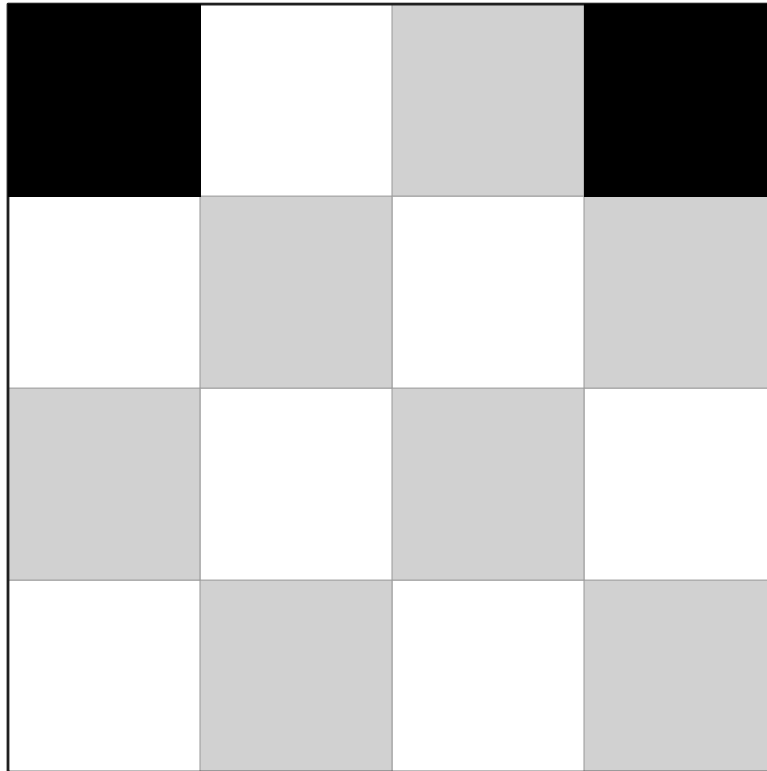
Domino 3 covers dark _____ and light _____

Problem 9: On this board two black squares are already drawn. Try to cover all the dark and light squares with dominoes. If you can, draw a check in the box. If you cannot, draw an X. Then count the dark squares that are not black and the light squares that are not black, and write the numbers.



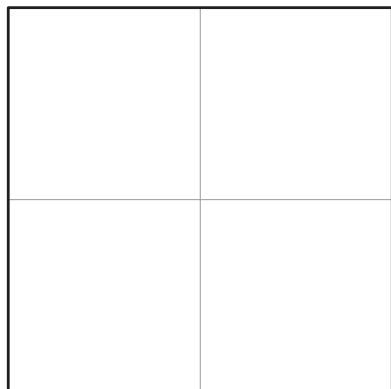
dark _____ light _____

Problem 9: (continued) Now the two black squares are on the same side. Try to cover all the dark and light squares with dominoes. Draw a check or an X in the box. Count the dark squares and the light squares that are not black, and write the numbers.

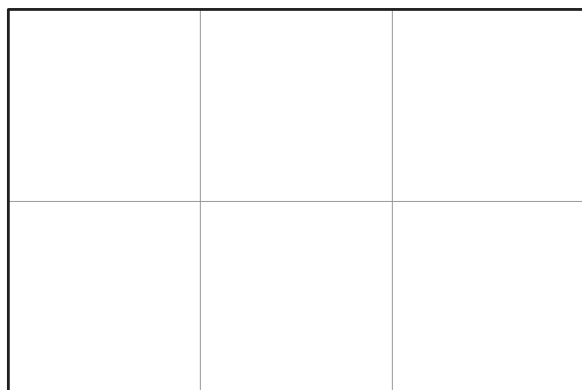
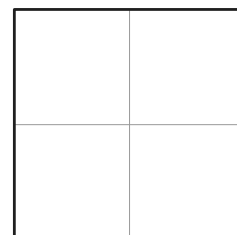
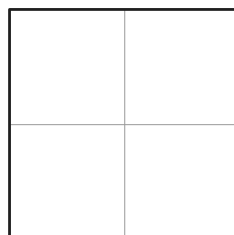
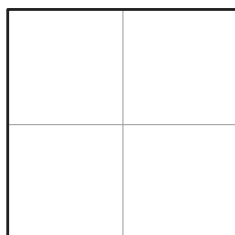


dark _____ light _____

Problem 10: Cover the 2-by-2 board with dominoes. Draw your way on a small board: draw a thick line around each domino. Find every different way, using a new small board for each. Then do the same for the 2-by-3 board.



2-by-2 board



2-by-3 board

